A small but consistent edge: forecasts tracked actual moves on all 97 unseen trading days

Can a model predict a stock's next 60-second move against the index in the Nasdaq closing auction?

About this analysis

The market. The closing auction sets each stock's official closing price, which funds use to value what they hold.

The data. 5.2 million order-book snapshots: 200 Nasdaq stocks, 481 trading days, the last 10 minutes of each day.

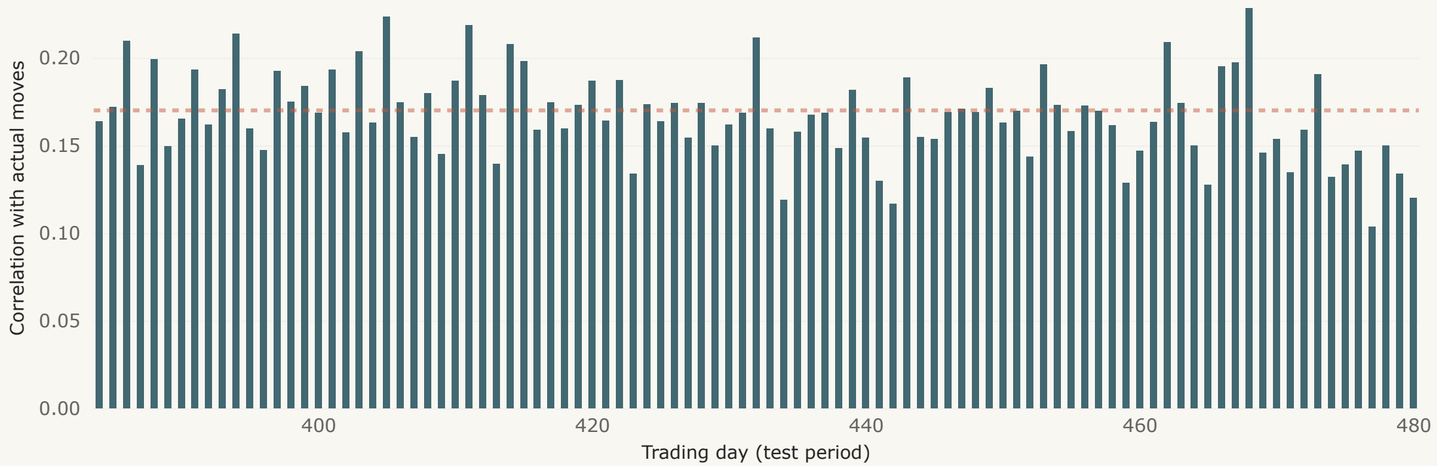
The model. LightGBM, trained on the first 384 days and tested on the last 97, which it never saw.

The target. How far each stock moves against the index in the next 60 seconds, in basis points (1 bp = 0.01%).

A baseline, not a competition model: top Kaggle entries cut error far more. Later pages show where it fails, what drives it, and how much to trust each prediction.

Correlation	Days Positive	Direction Correct	Out-of-sample R^2
0.17	97	54.9%	2.8%
0 = no skill	correlation above 0 every test day	50% = coin flip	small is normal for market returns

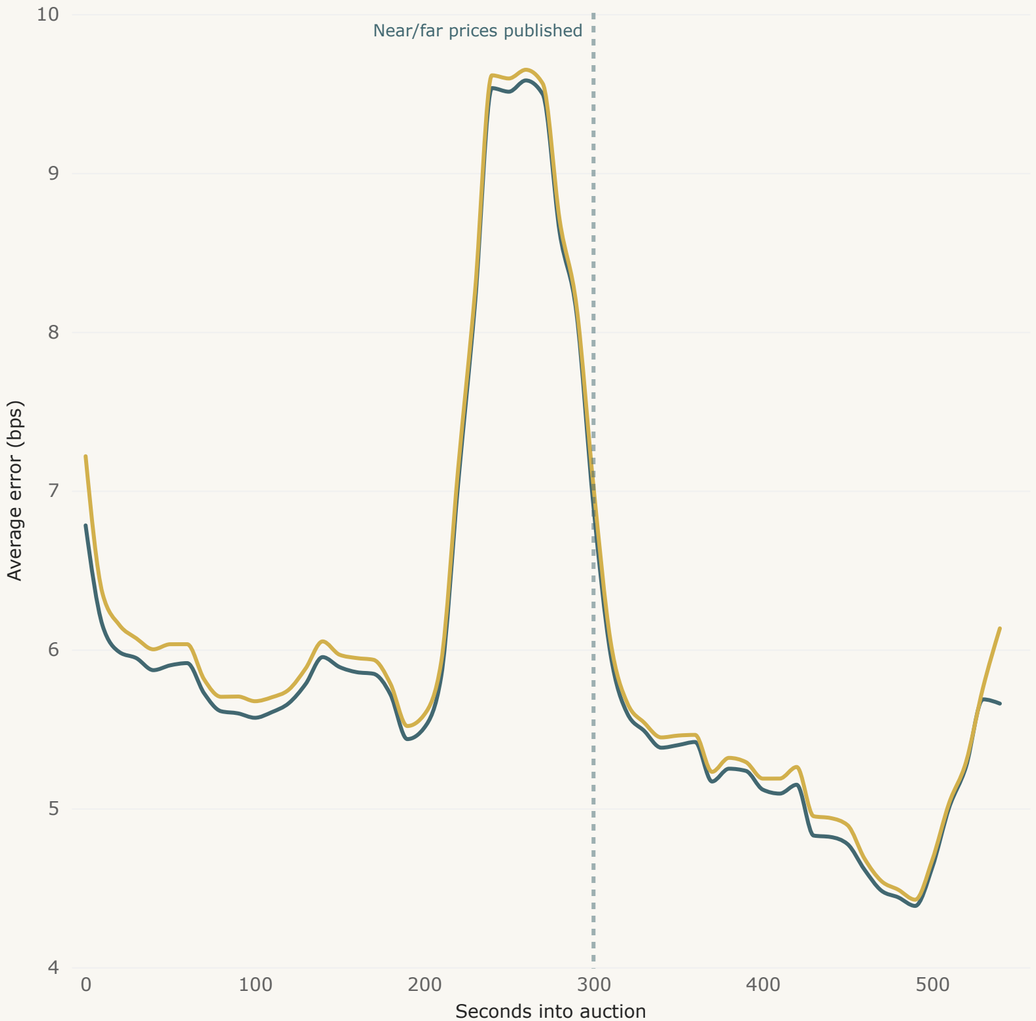
Daily correlation between forecast and actual move, test period (dashed line = average 0.17)



Riya Shet · Data: Optiver "Trading at the Close" (Kaggle)
Code: github.com/riyashet-hds/optiver-closing-auction-analysis

Where the baseline model fails

Errors spike just before near/far prices are published, and the model helps least there

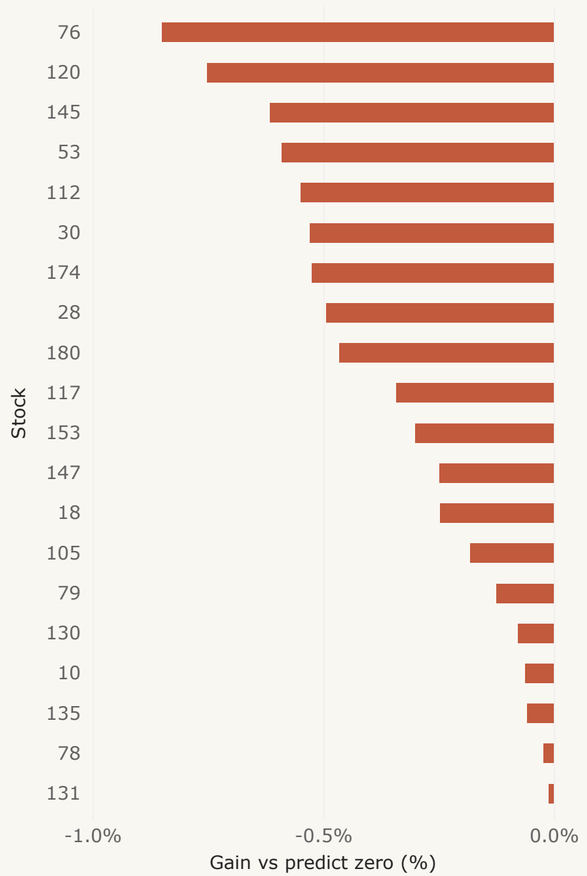


Model Error (bps)
5.96

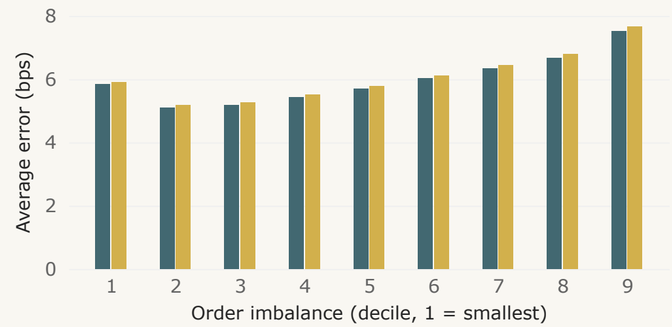
Predict-Zero Error (bps)
6.06

Improvement %
1.63%

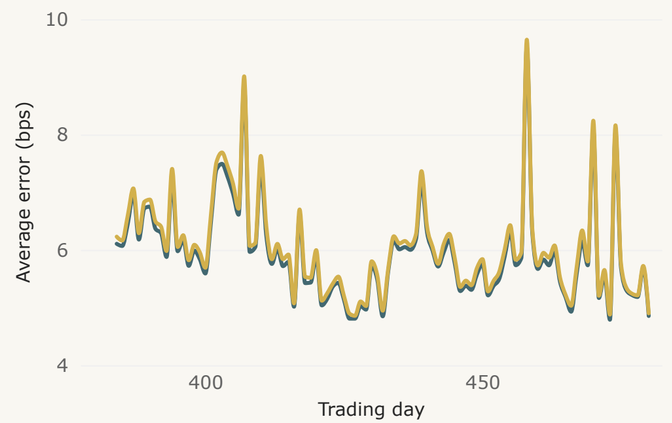
For 20 stocks, the model is worse than predicting zero



Bigger order imbalances, bigger errors



The gap stays small on every day

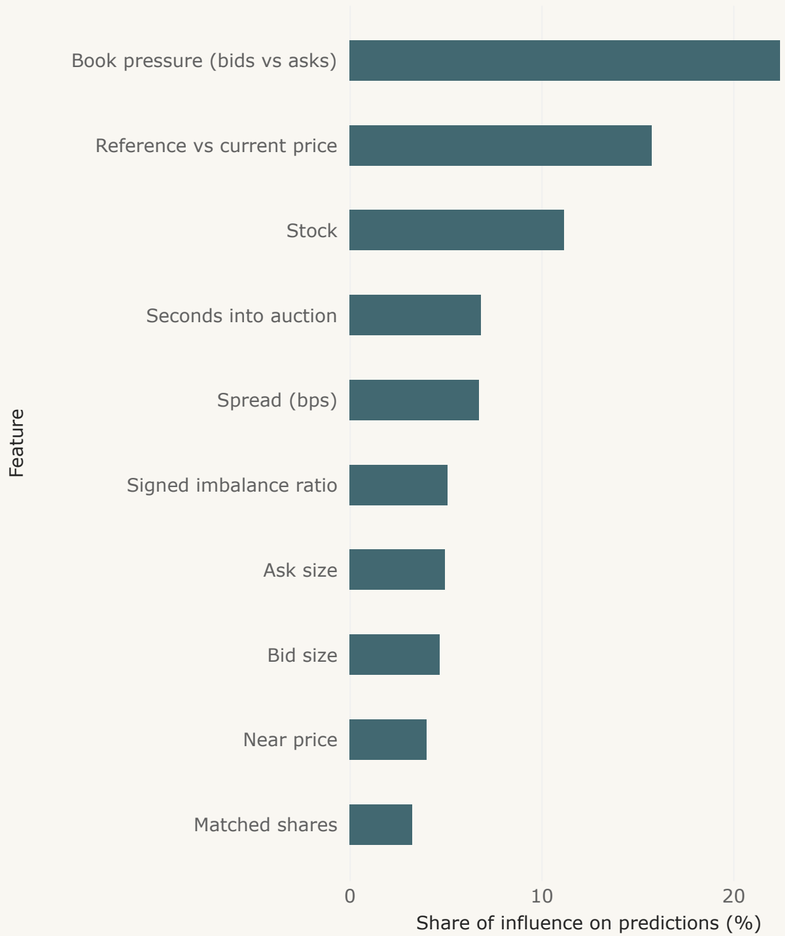


Why the model predicts what it does

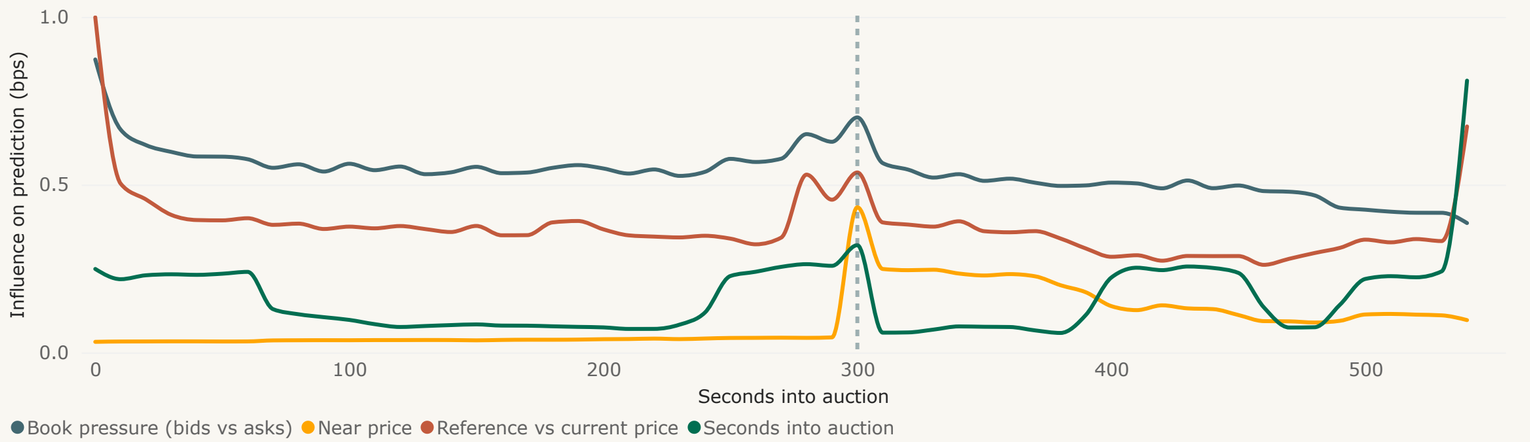
SHAP splits each prediction into how much every input pushed it up or down.

Two signals drive most predictions: bid-vs-ask pressure and the auction's reference price

Top 10 of 19 features; the other 9 add up to 15%.



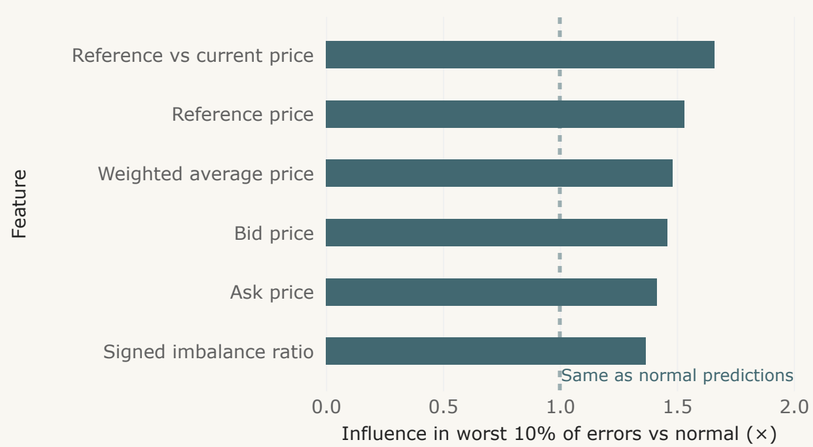
Near price becomes a key signal the moment it is published



When buyers dominate the book, the model expects the price to settle back down



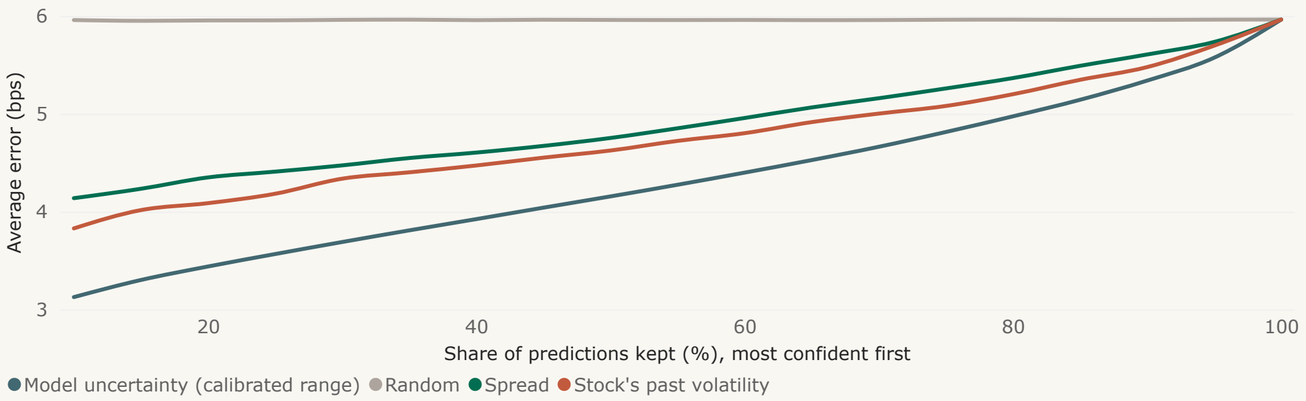
Big misses happen when the model leans on the reference-price gap



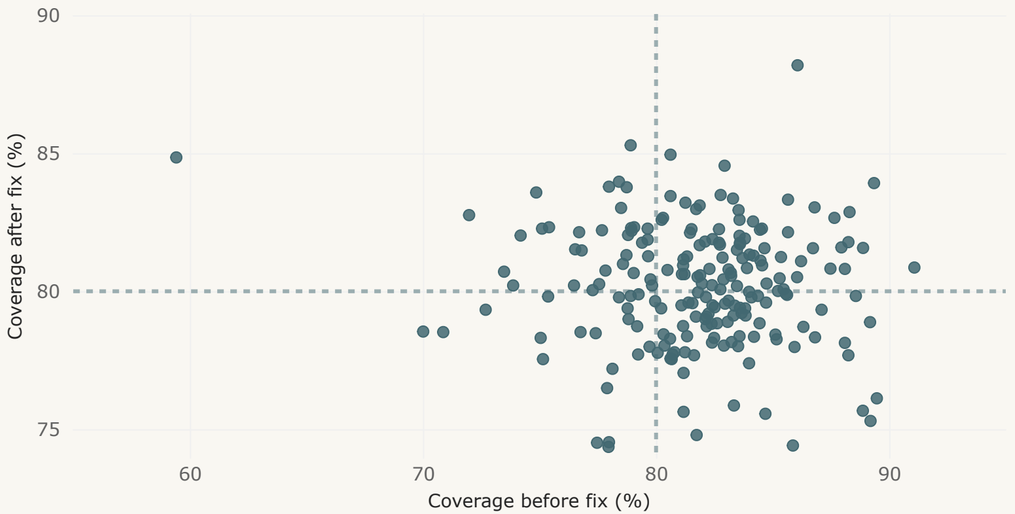
Sizing by confidence: how much to trust each prediction

The model gives an 80% range for every prediction. Coverage = how often the real value actually landed inside it.

Trading only on the most confident half cuts error by 30%, beating simpler rules

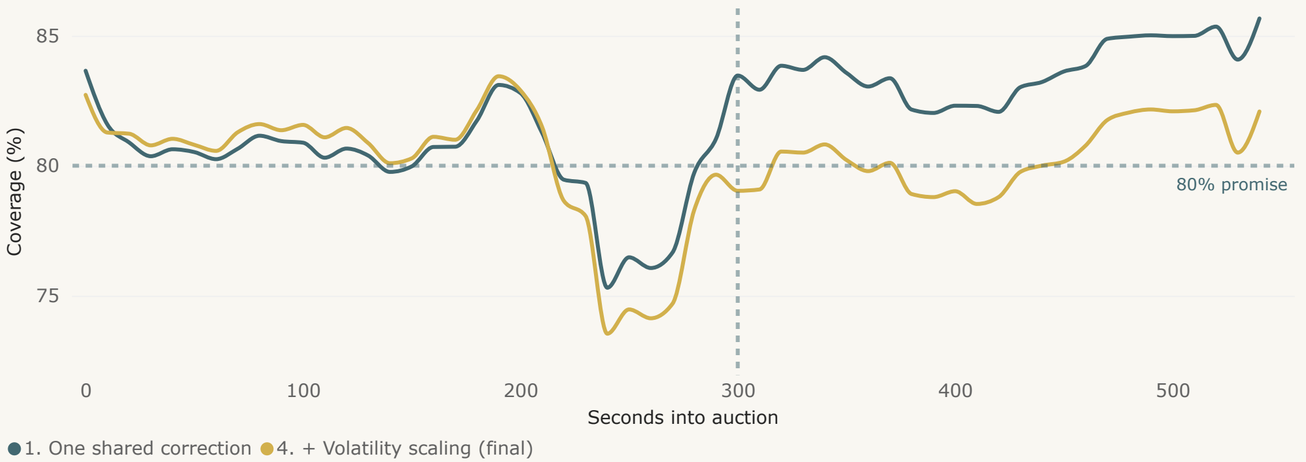


After the fix, stocks cluster closer to 80%: the old worst stock goes from 59% to 85%, and none falls below 74%



Coverage dips below the 80% promise just before near/far prices are published

The final method keeps the promise more evenly after 300 s, but not in the window before the release (known limitation).

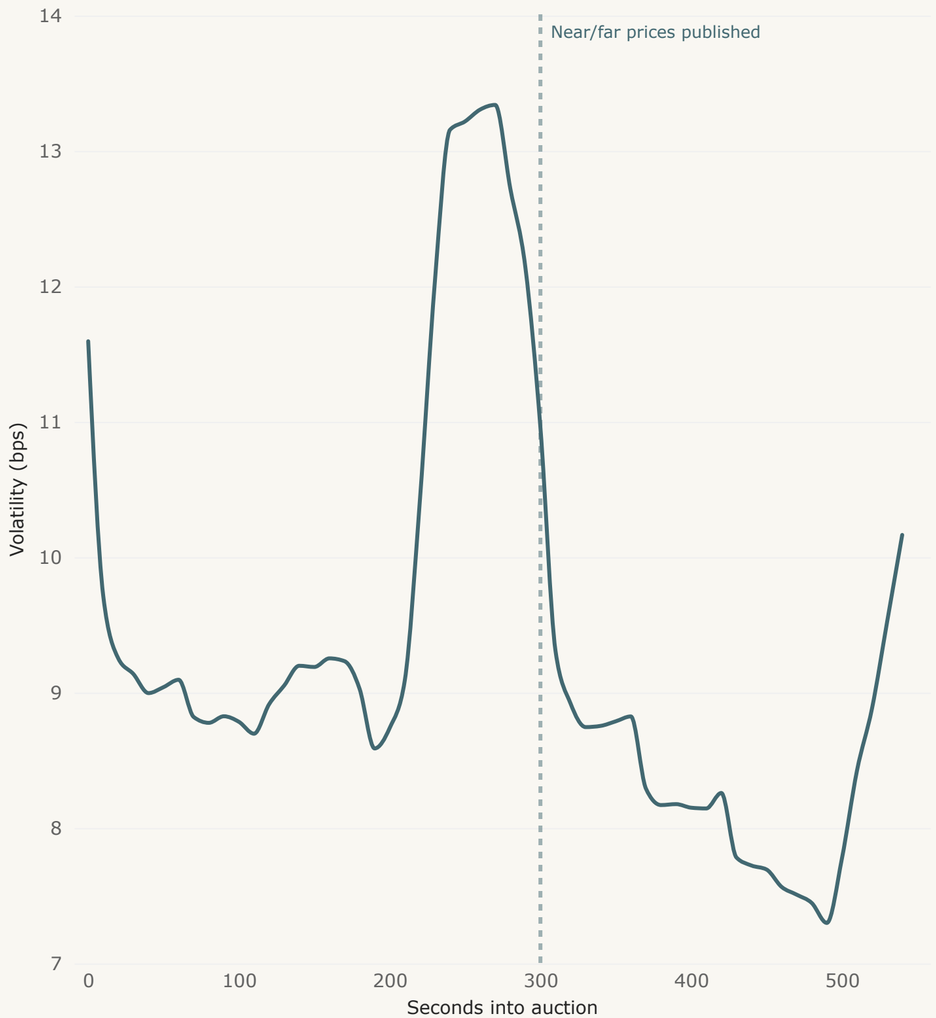


Each fix solved a different part of the problem

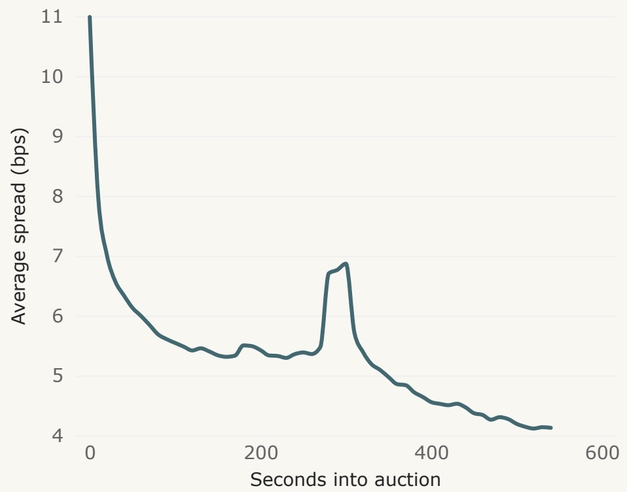
Method	Coverage (%)	Range width (bps)	Worst stock (%)	Stocks below 75%	Coverage at 220–290 s (%)
1. One shared correction	81.81	19.76	59.42	9	78.01
2. Per stock & phase	81.81	20.00	63.02	6	79.21
3. Per stock & phase + daily update	80.78	19.47	62.44	6	78.38
4. + Volatility scaling (final)	80.21	19.79	74.36	5	76.44

The Nasdaq closing auction: when prices are hardest to predict

Prices are hardest to predict just before near/far prices are published (300 s)



Spreads narrow as the auction progresses



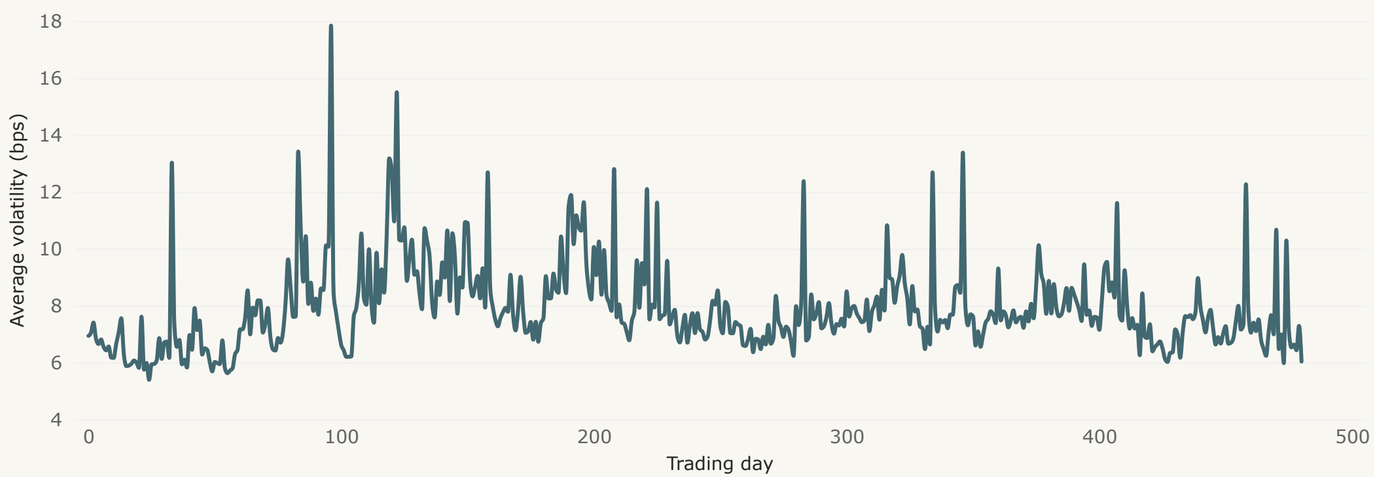
Stocks
200

Trading Days
481

Avg Spread (bps)
5.38

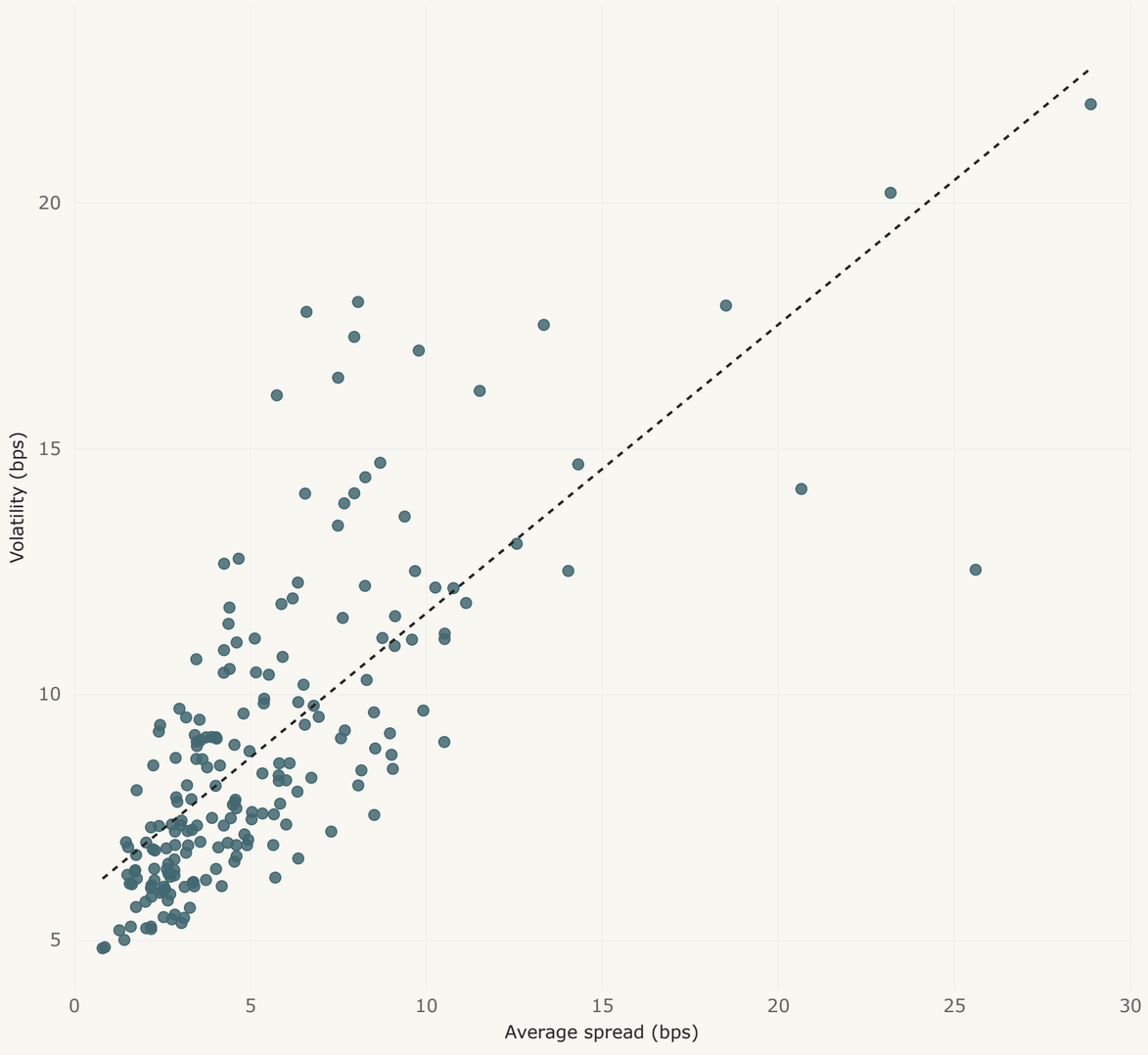
Avg Volatility (bps)
7.91

Volatility comes in bursts: a few days spike to 2–3× the norm

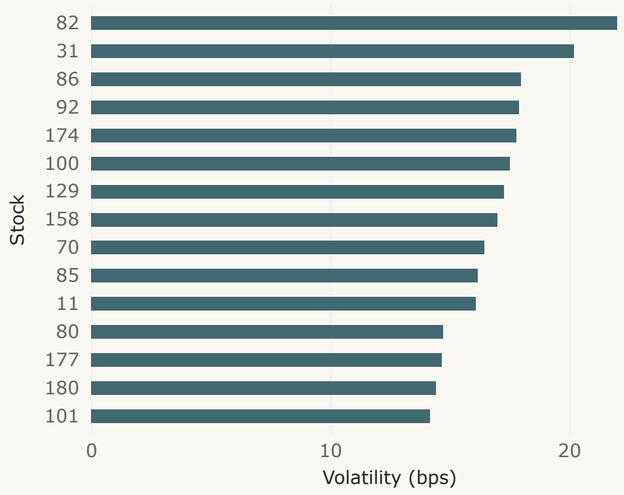


Which stocks are hardest to predict, and why

Wider spreads, harder to predict: thinly traded stocks are the noisiest



The 15 hardest stocks to predict



Hardest vs easiest stock (× volatility)

4.6

Spread–volatility correlation

0.75

(Pearson r across 200 stocks)

Stock	Difficulty rank	Volatility (bps)	Avg spread (bps)	Avg imbalance ratio
82	1	22.00	28.89	0.18
31	2	20.20	23.20	0.22
86	3	17.98	8.07	0.23
92	4	17.91	18.52	0.17
174	5	17.78	6.61	0.21
100	6	17.51	13.35	0.14
129	7	17.27	7.96	0.14
158	8	16.99	9.80	0.15
70	9	16.44	7.50	0.24
85	10	16.17	11.53	0.17
11	11	16.08	5.76	0.19
80	12	14.71	8.70	0.16
177	13	14.68	14.33	0.11
180	14	14.41	8.28	0.13